Bengali Dialect Identification for Analyzing the Geographical Origin of Speakers

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Dedicated to,

My beloved parents and sister.

DECLARATION

I certify that

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b. the work has not been submitted to any other institute for any degree or diploma.

c. I have followed the guidelines provided by the institute in writing the report.

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Abstract

Dialect detection systems are vital because they bridge the gap between standard AI models and real-world linguistic diversity. Through precise dialect detection, errors in transcription and understanding of spoken content are significantly minimized. Bengali is a linguistically diverse and widely spoken language across Bangladesh and a few parts of India, particularly in West Bengal and Tripura. Though the Bengali language is rich with dialectal variations, it is often ignored when it comes to building dialect detection systems for Automatic Speech Recognition(ASR). The current studies on the Bengali language have mostly focused on Standard Bangla rather than its dialectal variations. Also, there is no such research that considers both Indian and Bangladeshi dialects in a single study, to the best of our knowledge. This study bridged that gap by conducting experiments on dialectal Bengali for both India and Bangladesh.

Although several efforts have been made to categorize Bengali dialects in previous research, there is still a huge deficit in relation to the quality of the data used. More than 80% of current datasets were originally built for ASR or Text-to-Speech (TTS) systems, not for the dialect detection task. As an alternative solution to the quality constraint of current data, which cannot be solved by using human-annotated data due to scarcity, we considered the use of Self-supervised learning (SSL) with a neural network (NN) based classification head.

Keywords— Bengali Dialect Identification, Cross-border Linguistic Analysis, Low-resource Language Processing, Wav2vec 2.0 XLS-R, Speech Embeddings

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Introduction

1.1 Background

Dialectology is the study of regional variation in the phonology, vocabulary, grammar, and intonation of languages. Such variation is caused by geographical separation, migration, and social hierarchy, which create non-homogeneity within a "standard" language. Bengali, with over 300 million speakers worldwide, is said to have an extreme variation in dialect [1]. While only a few studies identify the variations in Bangladeshi Bengali dialects, even fewer studies are conducted to capture the dialectal variations of West Bengal [2].

Bengali Dialect Identification (BDI) is a speech processing task that deals with identifying special acoustic characteristics and speech patterns [3]. In most cases, Automatic Speech Recognition (ASR) systems fail due to dialectal variations[4]. Because phonetic changes affect speech patterns in a way that systems trained on standard language fail to capture those patterns. Changes in speech patterns create dialectal clusters where differences in tone and accent impose a challenge to global speech systems. In modern technology, this failure keeps a large native population deprived of AI welfare due to an accent barrier.

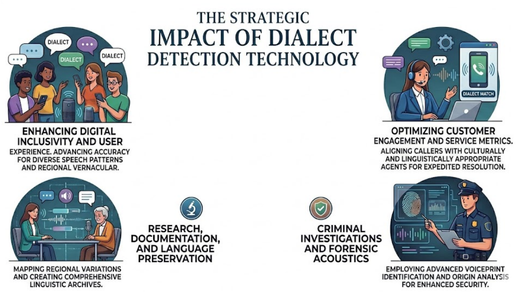


Figure 1.1: Application of dialect detection systems

1.2 Limitations

Current research on dialect detection in Bengali often ignores the substantial distinctions that arise within the linguistic structure of the languages used in other countries. The problem is that the insufficient quantity of quality corpora for the development of such applications persists to this day, especially concerning Bengali dialects of different Indian regions. Due to this deficiency of data, all recent works tend to be focused solely on the standard Bangla dialect, thus producing speech recognition systems unable to cope with the diversity of the Bengali dialects. To solve this issue and make technologies more inclusive, there should be an improvement in data quantity and diversity regarding various Bengali dialects.

1.3 Motivation

Correctly recognizing regional speech differences is crucial in the preservation of a language's diversity across locations and the improvement of digital services in rural areas. Aside from phonetic characteristics such as vowel lengthening, consonant deletion, and intonation patterns, these dialectal features can also serve as a non-invasive way to fine-tune speech recognition systems to non-standard speech[5]. Despite their efficiency in speech mapping, DNNs and self-supervised approaches have been considered **black boxes** because they do not show transparency in their classification processes.

For high-importance use cases like linguistic forensics and local education, explainability is key to ensuring that AI models can make predictions in any kind of situation, despite being a noisy border environment or a quiet classroom, even with low resources. As a result, this research aimed to assess and compare self-supervised models [6], to establish their effectiveness in processing Bengali speech, with Dialect Recognition models of other languages. Additionally, we carried out a post-hoc analysis to demystify the classification process by identifying the features that facilitate the differentiation in the dialectal speech samples.

1.4 Main Contribution

- This study bridged the gap of cross-border linguistics by considering both the Bangladeshi and Indian Bangla dialects.
- This study explored and addressed the significant limitations of the pre-existing data corpora for building a dialect detection system.
- This study experimented with SSL models to evaluate their performance on the dialects of the Bengali language.

1.5 Organization of the Thesis

In this dissertation, I built a deep learning-based model for dialect detection of low-resource Bengali dialects solely based on acoustic features. This report is structured into the following chapters:

- **Chapter 2** This chapter consists of a literature review on pre-existing data corpora and experiments conducted on a dialect detection system for both Bengali and other languages.
- **Chapter 3** This chapter provides a brief data description and the baseline model architecture, which is primarily used for the dialect detection task.
- **Chapter 4** This chapter consists the model improvement part by conducting a few experiments.
- **Chapter 5** This chapter provides a framework to improve the data by introducing data augmentation techniques.
- **Chapter 6** This final chapter presents the conclusions, limitations, and potential future directions of this study on Bengali dialect detection.

2

Literature Survey

Despite being a linguistically diverse language spoken across Bangladesh and a large part of India, it is a low-resource language. While there are various research works using standard Bengali, unique regional variations are often ignored, especially the dialects of West Bengal. Building an accurate Bengali dialectal audio corpus is crucial for preserving linguistic heritage, enabling targeted socio-economic services in rural areas, and supporting border security in the current tense situation between the two countries.

2.1 Related works on dialect detection

The story of dialect identification begins with a shift in the way that machines process audio signals. For decades, researchers have used rigid statistical methods such as Gaussian Mixture Models (GMM) and i-vectors. These methods have proven to be effective, yet lacked the complexity to grasp the emotion and rhythm of human language. This period of time has been characterized by an attempt to find mathematical certainty within a medium as fluid as human language. The revolution began with the invention of Self-Supervised Learning (SSL) techniques. Models such as Wav2Vec 2.0 (XLS-R 300M) [7] have fundamentally changed the story by training on nearly half a million hours of raw audio data from 128 languages. Rather than memorizing words, it discovered a "universal phonemic space." This allowed it to detect the subtle internal variations of even rare dialects such as those spoken within the North Sámi regions. While Hidden-Unit Bidirectional Encoder Representations from Transformers (HuBERT)[8] ushered in a new era of resilience by training models to ignore background noise through discrete pseudo-labeling, the Emphasized Channel Attention, Propagation, and Aggregation in Time Delay Neural Network(ECAPA-TDNN)[8] emerged as the specialist, using "attention" to zoom in on specific frequencies that define regional identity. This marks a global shift from basic sound recognition to a deep understanding of the human voice.

Table 2.1: Summary of significant previous works in dialect and speech identification

Research Name	**Author(s)**	**Short Description**
Wav2Vec 2.0 (XLS-R) [7]	Babu et al.	A large-scale cross-lingual model trained on 128 languages for universal speech representations.
HuBERT [8]	Hsu et al.	Utilizes masked prediction of hidden units to learn robust features from raw audio data.
ECAPA-TDNN [8]	Desplanques et al.	Employs channel attention and multi-layer aggregation to focus on identity-defining frequencies.
BanglaDial Dataset [9]	Islam et al.	A comprehensive study providing a large-scale dataset for 11 regional Bengali dialects.
Dialect Identification via GMM-UBM [10]	Torres-Carrasquillo	An early influential work using statistical models to distinguish between closely related dialects.

2.2 Research in Dialect Identification on Languages of India

India is home to some of the most linguistically diverse peoples anywhere in the world. For a machine learning system, this represents a "Data Scarcity" Paradox. While languages such as Hindi or Tamil may have large amounts of training data, their dialects may not. Researchers have traditionally addressed this problem through a move away from simple acoustic analysis to Transfer Learning. The story of Indian dialect detection is one of adaptation – of teaching the global rhythms to the local heartbeat. A large part of the more recent Indian research has been the struggle with code-switching – the blending of English[11] with regional dialects – and the prosodic change between urban and rural residents. For example, Microsoft's Respin project, as well as several projects led by the IITs, have shown that a model may be capable of understanding "Standard" language, but will flounder in the face of the heavy phonetic changes that occur between the lips of rural barriers.

Bengali is the second most spoken language in India and the national language of Bangladesh, yet for years, the digital research was isolated by national borders. For example, in Bangladesh, researchers are focusing on the most unique, independent-sounding local languages—like those in Chittagong (Chatgaya) and Sylhet[12]. These are so different in tone and sound that they are almost like separate languages, not just accents of standard Bengali. In West Bengal, India, research has focused on the "Rarhi" or "Manbhumi" accents[13]. The "gap" in this narrative is that, until now, a model trained on Bangladeshi dialects would have difficulty with Indian Bengali nuances, and vice versa. But this research is a turning point in this narrative. By drawing on the XLS-R 300M, which has already been trained on the phonemes of 128 languages, we are finally building a cross-border bridge.

3

Dataset Design and Baseline System

To the best of our knowledge, there are no studies on dialects of the Bengali language that include speakers from both Bangladesh and India. If choosing different datasets for both countries, it is necessary to guarantee equality regarding the nature of the data (both should be collected spontaneously) and the socio-economic status of the speakers. This condition is vital since one's environment affects speaking style.

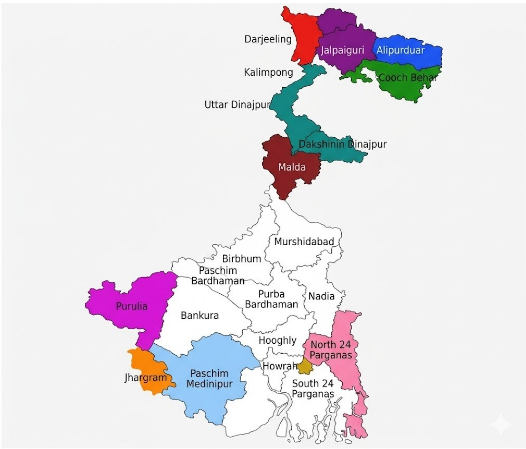
3.1 Datasets

For this research, we have chosen Project Vanni for the Indian Bangla dialects and RBVD[14] (Regional Bangla Voice Dataset) for the Bangladeshi Bangla dialects. Since only project Vaani is the only study that preserved dialects of the Bengali language of India with the most precision, we considered this as our Indian set. On the other hand, the only open source, authentic dialectal data and the most comprehensive Bangladeshi dialect can be found only in the RBVD dataset. Also, the RBVD dataset complements the Vaani project the most.

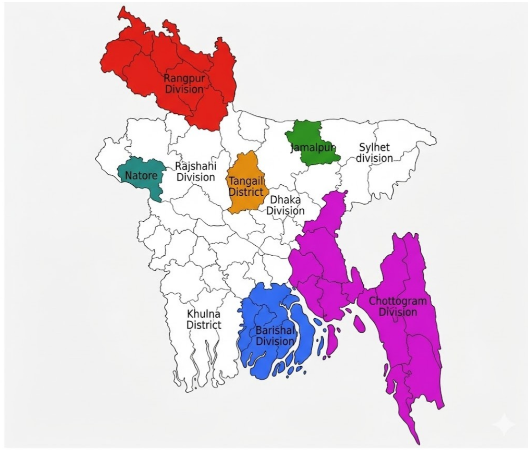
Table 3.1: Comparison of dataset properties

Feature	**Project Vaani (India)**	**RBVD (Bangladesh)**
Speech Type	Read Speech (Image-based)	Read Speech / Semi-structured
Total Duration	~8.5 Hours (WB subset)	~1 Hour
Regions Covered	11 Districts of West Bengal	8 Districts of Bangladesh
Sampling Rate	16 kHz / 48 kHz	16 kHz / 44.1 kHz
Gender Ratio	~52% Female / 48% Male	Not-Specified

Comparing the two datasets, the similarities of their structure and technical parameters reveal a striking level of homogeneity, a necessity for an equitable cross-border dialect research. The presence of Read Speech predominantly in both sets provides the phonetic foundation for the model. Moreover, the natural linguistic variability introduced by semi-structured texts in the RBVD set serves as an ideal complement to the photographic stimuli used by Project Vaani. From the geographical perspective, the datasets contain samples from a wide range of regions: 11 districts of West Bengal and 8 districts of Bangladesh, ensuring exposure to various sub-dialects other than standardized urban variants. Importantly, both sets are compatible with the 16 kHz sampling rate required by the XLS-R framework, preventing any frequency-related disparities. While Project Vaani demonstrates remarkable gender diversity, represented in equal proportions of females (52



(a) Regions in West Bengal



(b) Regions in Bangladesh

Figure 3.1: Geographical regions covered in the study datasets.

The RBVD dataset only contains around 1,067 samples. To keep the datasets balanced, the same number of samples was drawn from the Vaani data. Then it was fed to the preprocessing pipeline.

3.2 Audio Preprocessing

Before feeding our audio data to the XLS-R 300m model, we performed several standardization steps to ensure the training process is stable and the model provides satisfactory results. First, all the audio files were converted into .wav format, and then they were passed through a preprocessing pipeline given below. During preprocessing, corrupted .mp3 files caused MPEG header error. To handle that, automated exception handling was implemented, which led to a reduction in the number of samples in the RBVD dataset.

3.2.1 Signal Standardization

The first step involved the normalization of the raw data into a consistent digital format. This ensures that variations in recording equipment do not bias the model. Since the source datasets contain different sampling rates, all files were resampled to 16 kHz, the required frequency for the XLS-R architecture. Additionally, the audio was converted to mono to eliminate redundant spatial information. Finally, the signal was standardized to 16-bit PCM and peak-normalized to 0 dB to ensure uniform loudness across the corpus.

$$x_{norm}[n] = \frac{\mathcal{R}_{16k}\left(\frac{1}{C}\sum_{c=1}^{C} x_{raw,c}[n]\right)}{\max_m \left|\mathcal{R}_{16k}\left(\frac{1}{C}\sum_{c=1}^{C} x_{raw,c}[m]\right)\right|} \tag{3.1}$$

- $x_{norm}[n]$ represents the final normalized discrete-time signal.
- $x_{raw,c}[n]$ is the original input signal for the c-th channel.
- C denotes the total number of audio channels (averaged to convert the signal to mono).
- $\mathcal{R}_{16k}(\cdot)$ represents the resampling operator that converts the audio to a sampling rate of 16 kHz.
- n and m are discrete-time indices.
- The denominator $\max_m |\cdot|$ denotes the peak absolute amplitude, used to scale the signal to a maximum of 0 dB.

3.2.2 Voice Activity Detection (VAD)

To eliminate non-speech segments and background silence that could misguide the model, we have used the SileroVAD model[15]. The code identifies human speech segments and removes long periods of silence or background noise. Only human speech chunks were selected and concatenated to yield a condensed audio segment of pure vocals.

3.2.3 Signal-to-Noise Ratio (SNR) Filtering

In order to maintain data quality, we have filtered the samples by SNR[16]. The noise was estimated based on the 5 % least energy samples in a clip. Clips with a low SNR below 15 dB are discarded. This ensures that our model is not learning to map certain dialects to low-quality audio or ambient noise. To ensure data quality, we filter the samples based on the signal-to-noise ratio (SNR) defined as:

$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10} \left(\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}} \right) \tag{3.2}$$

where P_{signal} is the average power of the clip and P_{noise} is the noise power estimated from the bottom 5% energy samples. Clips are retained only if $\mathrm{SNR}_{\mathrm{dB}} > 15$ dB to prevent the model from associating specific dialects with background noise.

3.2.4 Fixed-Length Transformation

Deep learning networks, especially Transformers, perform best when all dimensions in the input are equal. All samples were normalized to a standard length of 5 seconds. If samples were longer than 5 seconds, only the initial "high information segment" is kept. If samples were shorter than 5 seconds, then instead of padding them with silence, the audio was repeated to meet the standard duration.

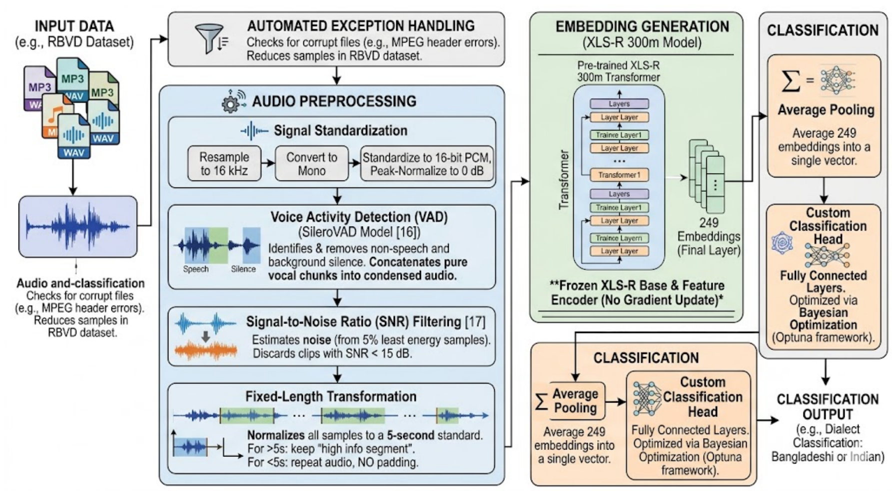


Figure 3.2: Baseline model architecture

After preprocessing, the Bangladeshi set was left with 824 samples, and the

Indian set was left with 825 samples, which were then split into train and test sets. The train category had approximately 660 samples for the train set, and the rest were the test set.

3.3 Baseline Model

The data was split into train and test (80/20) sets, using a random seed of 42 for consistent distribution of data. Then it was trained on a Kaggle environment using Nvidia Tesla T4 GPUs. Since we have a small data corpus (only 825 samples) for each Bangladeshi and Indian segment, we didn't fine-tune the whole model. The base XLS-R feature encoder was strictly frozen with no gradient update, and a custom classification head was added to it. For the baseline model, embeddings were generated from the final layer. After performing average pooling on the 249 embeddings generated by the XLS-R, we fed them to the fully connected layers. The fully connected layers were optimized using a Bayesian Optimization strategy using the Optuna framework.

Table 3.2: Optimal hyperparameters for the fully connected layers

Hyperparameter	**Value**
Optimization Framework	Optuna (Bayesian)
Batch Size	8
Gradient Accumulation Steps	2
Learning Rate	4.25e-05
Weight Decay	0.0001
Hidden Dimension	1024
Activation Function	LeakyReLU
Epochs	45

The performance of the baseline model is given below.

Table 3.3: Classification report of base-line model

Class	**Precision**	**Recall**	**F1-score**	**Support**
bn_bd (Bangladesh)	0.84	0.91	0.87	164
bn_in (India)	0.90	0.83	0.86	165
Accuracy			0.87	329

The initial model was able to reach a balanced accuracy of 87%, proving that the use of pre-trained XLS-R embeddings has been able to adequately encode the regional phonetics of the language without fine-tuning. Although the model can accurately recognize Bangladeshi Bengali speakers with a recall of 0.91, meaning it accurately predicted whether the speaker is Bangladeshi Bengali, it can more accurately predict the class Indian Bengali due to the higher precision of 0.90, meaning that it produced fewer false predictions for the Indian Bengali class.

A similar kind of architecture can be found for North Sami Dialect Identification and Arabic Speech Dialect Identification; these two studies. The comparison of them in Table 3.4.

Table 3.4: Comparative accuracy of XLS-R 300m across languages

Language / Dialect	**Model Architecture**	**Accuracy (%)**
Arabic	XLS-R 300m E2E	76.20
North Sami	XLS-R 300m Classifier	95.00
Bengali (BD vs. IN)	XLS-R 300m Classifier (Ours)	87.00

We can see that our baseline model significantly outperforms the Arabic language[8], but couldn't outperform the North Sami dialect detection system[6]. This results in two aspects of the project being divided, one is improving the model to give better performance, and the other is to introduce variations in the data, allowing the model to generalize better in real world scenario. Though we can make the above inferences, this comparison must not be taken seriously as a benchmark, since the data size and quality were different for those studies.

One of the main drawbacks of the baseline architecture is the strict freezing of the feature extractor. Since the underlying XLS-R model does not receive any gradient updates during training, the fundamental acoustics are unable to adjust to the distinctive sounds of Bengali dialects. Consequently, the design is constrained to use only the general, pre-trained embeddings. This poses a considerable challenge to the model, as the limited classification layer is required to do all the hard work of separating dialects without the help of the fine-tuned feature extractor. To overcome this problem, experiments were conducted in the later sections.

4

Model Architecture Advancement

It has been shown that self-supervised models like XLS-R retain a hierarchical structure for their transformer layers[7]. The first 8 layers retain low-level features like pitch and gender. In these layers, speaker separation is most distinct. Layers 9-16 retain maximum phonetic information. Cues for discriminating language are maximum in these layers. The rest of the layers (17-24) retain contextual meaning. The question remained whether giving equal importance to each layer affects the model's performance or not. We have used this information to initialize a new model architecture.

This new model design consisted of a weighted layer pooling classifier with a Wav2Vec2 XLS-R (300M) pre-trained model serving as its base structure. To incorporate a wide range of features, the classifier utilized the weight coefficients for each of the 25 transformer layers (i.e., 24 encoder layers, plus one embedding layer), where the initialization of weight coefficients was set to a uniform vector of ones, meaning that each layer initially has the same contribution ratio of 4% towards the overall output representation. In the forward propagation process, the normalized weights were calculated by using the Softmax function, which served to aggregate the outputs of all hidden states into a weighted vector that undergoes mean pooling along the sequence dimension before being fed into an MLP classifier head. The weights assigned to each layer output were trainable, and the MLP classification head was also optimized via the Bayesian optimization technique, as in the previous experiment.

We have gone through the same preprocessing steps to prepare our data for training, and the results are as follows.

This updated version has an accuracy rate of 81% compared to the base version, and this is slightly worse than the previous one but it shows a marked change

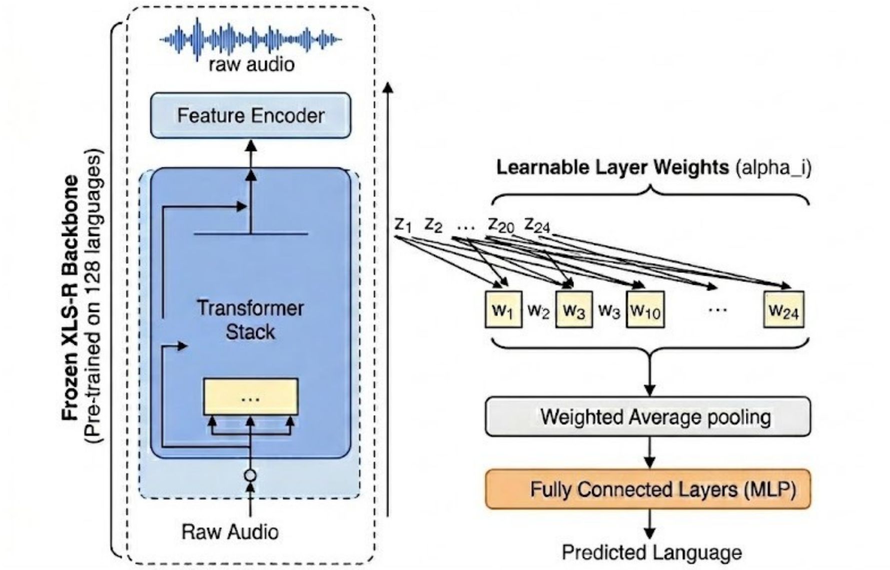


Figure 4.1: Weighted model architecture

Table 4.1: Classification report of improved model

Class	Precision	Recall	F1-Score	Support
bn_bd (Bangladesh)	0.73	1.00	0.84	164
bn_in (India)	1.00	0.62	0.77	165
Accuracy			**0.81**	**329**

in the behavior of classification. In addition, the updated model has achieved a 100% recall rate for Bangladeshi dialects, meaning that all Bangladeshi samples were correctly classified, but this comes at the expense of having a very low precision level (73%) because of the wrong classification of many Indian samples. On the other hand, the recall rate of Indian dialect samples is quite low (62%), but precision was maximized (100%).

4.1 Comparison of the Two Models

4.1.1 Understanding Layer Importance

From figure 4.2, it was observed that although we introduced the weighted averaging on the embeddings from each layers the most important contributions were made by the final layers for both classes(the 23rd layer is the most important one).

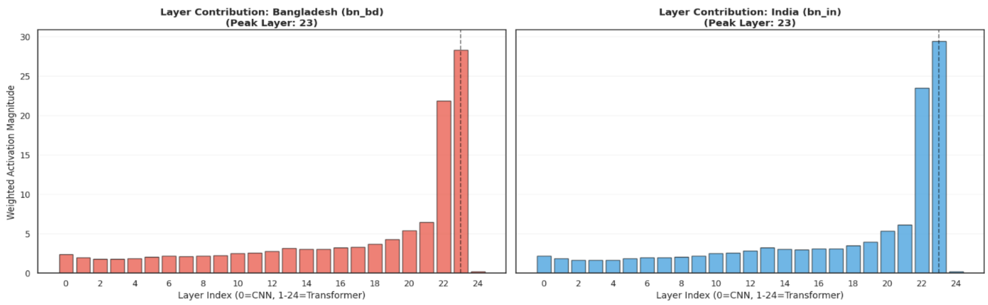


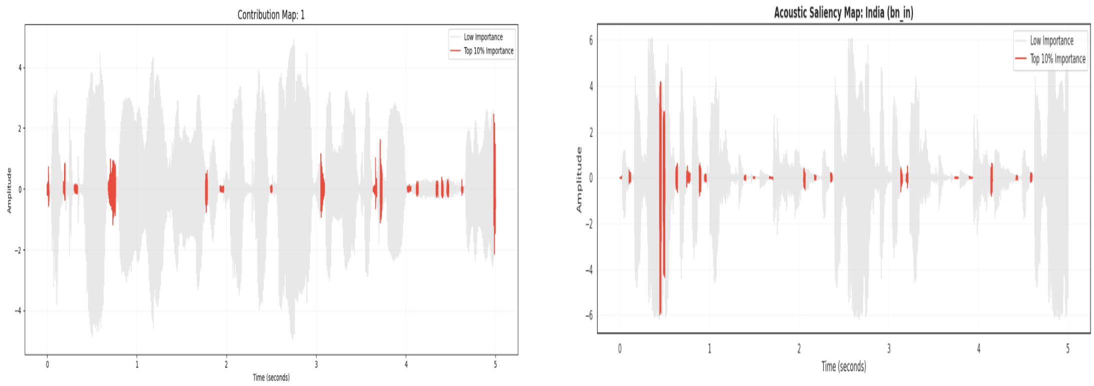
Figure 4.2: Model layer importance comparison for both dialects

In XLS-R, it is common for the intermediate layers (layers 9 to 16) to represent linguistic and acoustic properties relevant for accent detection, while the high-level layers (layers 17 to 24) represent semantic information. Figure 4.2 clearly shows that the model leans heavily toward the use of Layer 23. The implication is clear - there is something seriously wrong, as the classifier is likely using lexical information to distinguish the two data sets, as opposed to dialectal pronunciation differences. This might be caused by different sources of audio data, which had a different approach to collecting data. Also, we don't have the knowledge from which segment of society the RBVD data was collected from. Hence, there might be a misalignment of the two datasets.

4.1.2 Understanding Most Important Part of the Audio

In order to find the most influential parts of the audio, the following approach was used. At first, we extracted the attention weight matrix produced by the last layer of the Transformer architecture. In order to get the importance consensus, we calculated the average of weights across all attention heads and sum them up for each audio token separately; the higher the sum, the more the model depends on this specific audio part during classification. Then, we applied the 90th percentile threshold and select the top 10 percent audio tokens based on their summed weights. Finally, in order to pinpoint the timestamps on the actual waveform, we calculated the time ratio of the length of the tokenized audio input and the raw audio array length.

From Figure 4.3, we can see that both models classified the audio category by imposing importance on a smaller region rather than the full audio file. Hence,



(a) Audio Importance for Baseline Model (b) Audio Importance for Improved Model

Figure 4.3: Side-by-side comparison of dialect classification results.

we can say that rather than capturing the rythm the model is focusing on vowel transitions and consonant bursts. This makes the model vulnerable to learning shortcuts based on background noise or speaker identity, not the actual regional accent. Moreover, dialects exhibit prosodic characteristics such as rhythm and intonation, which cover longer periods of speech. The extremely sparse distribution in the salience maps (Figure 4.2) shows that the network completely disregards prosodic information at the higher temporal resolution scale. On the contrary, it is highly sensitive to small fluctuations lasting milliseconds, which could imply that it is prone to overfitting the particularities of the recording noise.

On the other hand, it can also be said that there might be a lack of generalization to capture the overall rhythm and prosody of the audio. To address this gap, we implemented various data augmentation techniques in the next chapter. This might solve two main problems. First, a low amount of data, and hopefully, it will introduce more generalization to the model.

5

Data Augmentation

In the previous chapter, we saw that even after improving the model architecture, the most important layers are still the final few layers. And also, the model is making its decision based on a few selective parts, not based on the whole rhythm of the audio. In the previous chapter, we also suspected that there might be a lack of generalization in the data. To overcome this problem in this chapter, we augmented the data. This will solve the lack of generalization problem, and we will get more data for the model to be trained on.

5.1 Data Augmentation Techniques

Data augmentation is a technique used to artificially increase the size and diversity of a training dataset by creating modified versions of existing data. To augment our data, we took the help of the built-in functions of the "audiomentations" Python library[17]. Unlike common practices where the test data remains untouched by synthetic manipulations, the augmentation process used in this study was conducted on full data and then splitted in to train and test sets. The intention was that the final model would classify the highly manipulated acoustic distribution, including artificial background sounds and pitch changes.

- **Add Background Noise:** All the files were subjected to the introduction of environmental sounds such as rainfall, dog bark, laughter, wind noise, and breathing sound. To simulate these distortions, we applied the ESC-50[18] dataset, where the distortion was applied with the Signal-to-Noise ratio (SNR) value randomly drawn from the range of 5 dB to 15 dB.
- **Pitch Changing:** Considering that it is important to maintain the characteristics of the original audio files in terms of distortion, it was not possible

to conduct global pitch alterations on all the files. Rather, we divided our database into two parts; one part consisted of 400 files, where a pitch up alteration was made by +2 to +4 semitones, whereas the rest of the files had a pitch change of -4 to -2 semitones.

- **Time Stretch:** This strategy was employed once again to extract the variation in the speech speed and drawls. In order to do this, a selected 500 recordings were scaled using varying speeds ranging from 0.8x slower to 1.2x faster. In order to maintain consistent input size for the neural network and have a standardized output size, all the outputs were normalized to be 5 seconds (80,000 samples @ 16kHz).

After augmenting the data, we ended up with approximately 2400 samples for each dialect class, which is 3 times the original data. Now, to see if augmentation helps improve our model performance or not, we trained our improved model.

5.2 Model Performance Analysis

Table 5.1: Comparative performance of baseline, improved, and data augmented models

Model Stage	Class	Precision	Recall	F1-score	Accuracy
Baseline	bn_bd	0.84	0.91	0.87	**0.87**
(N=329)	bn_in	0.90	0.83	0.86	
Improved	bn_bd	0.73	1.00	0.84	**0.81**
(N=329)	bn_in	1.00	0.62	0.77	
Augmented	bn_bd	0.70	1.00	0.82	**0.78**
(N=1190)	bn_in	1.00	0.56	0.72	

The Augmented model thus reflected the transition from a focus on laboratory accuracy towards practical effectiveness in achieving an average accuracy rate of 78% in a database about three times larger than used in the first experiments. The key characteristics of the model under analysis are represented in an unbalanced perfection in terms of Recall (1.00 for Bangladeshi), where not a single speaker of this dialect will be overlooked, and Precision (1.00 for Indian), where every positive prediction regarding India is completely correct. Although both the Baseline model

(87%) and the Improved model (81%) have higher overall accuracy rates, their experiments were conducted on significantly smaller databases with fewer acoustic variations. As a result of data augmentation, the new model sacrifices part of its ability to recall Indians (0.56) in favor of high reliability when dealing with varying sound environments and spontaneous speech.

Previously, we had suspected that the model performance is low due lack of generalization. But this experiment clearly shows that the problem lies elsewhere. Figure 5.1 shows that the model still captures local verb distortions rather than the full rhythm of the audio.

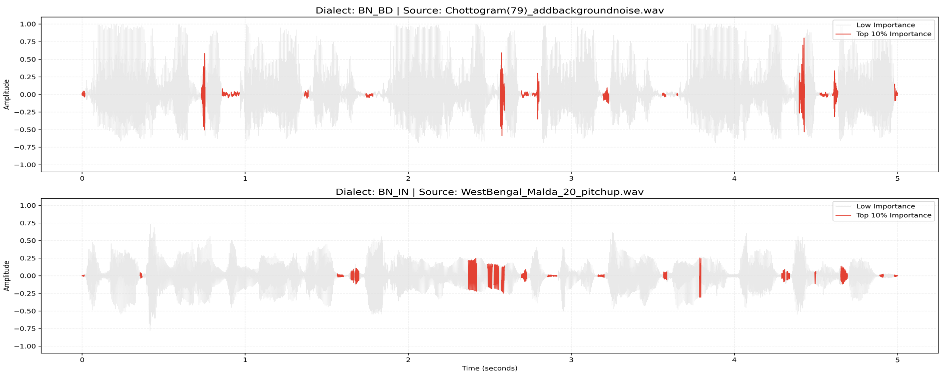


Figure 5.1: Contribution map of audio files.

It should be noted that the purpose of adding ESC-50 noise was to function as a regularizer that would compel the model to pay no attention to the environmental artifacts from the recording process and consider only the continuous dialect prosody. Yet, given the extreme sparseness and the presence of the spike-like features in the augmented saliency map (see Figure 5.1), the role of this kind of noise can be considered rather distracting.

Furthermore, the analysis in Figure 5.2 shows that there is an almost uniform distribution of layer weights, such that each layer out of the 24 layers has a weight of about 4 %. While this does not show equal informational value for all layers, it shows that the optimization procedure has failed to converge. The use of noise and different pitches may have made it difficult for the model to recognize the gradient pattern that was present in the previous exercise, which made Layer 23 dominant. This made the model unable to change focus and move to the middle layers, hence the uniform weights.

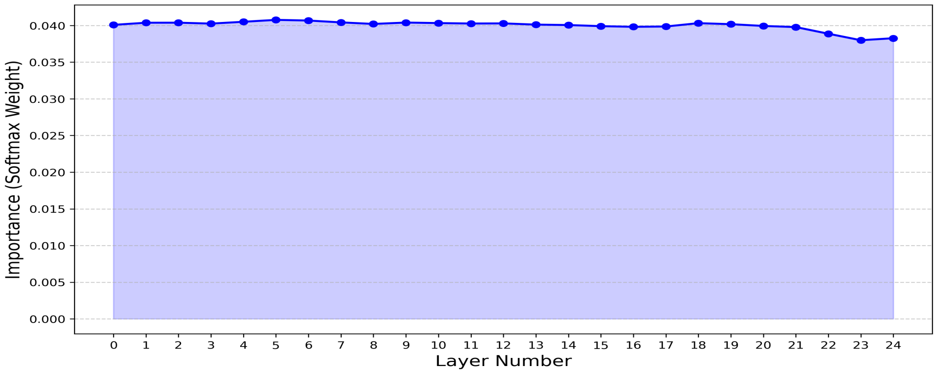


Figure 5.2: Layer importance graph

6

Conclusion

6.1 Conclusion

This research effectively bridged the cross-border gap for Bengali Dialect Identification by assessing the efficiency of the self-supervised XLS-R 300M model on dialects from Bangladesh and India. In order to tackle the problem of extreme data scarcity and test the generalizability of the model, the research adopted a baseline frozen-encoder framework, a sophisticated layer-weighted structure, and several data augmentation methods like pitch shifting, time stretching, and adding background noise. Despite its promising performance with 87% accuracy, post hoc analysis identified a significant structural weakness in the baseline model as it overwhelmingly weighted the 23rd transformer layer, suggesting the dependence of the model on semantic lexical features rather than phonetic dialectal differences. Moreover, acoustic saliency maps showed that the model mainly concentrated on individual acoustic events, including vowel onset and consonant burst, but ignored the prosodic rhythm found in natural dialects. Nonetheless, data augmentation tripled the training data size and made the model adapt to different sound environments, resulting in 78% accuracy that can be considered much more practical in real-life applications.

Despite of these issues, however, the study clearly demonstrated the critical gaps in Bengali speech processing that need to be addressed. By identifying how transformers may easily fail due to bias in datasets and environmental noise, the study created a technical guide for future studies on the issue.

6.2 Limitations and Future Directions

There are several limitations in this project regarding data availability, model architecture, and evaluation techniques, which simultaneously provide clear avenues for future research and improvement.

- **Data Scarcity and Corpus Standardization:** The initial difficulty relates to the problem of scarcity of data and unbalance issues. Although there is plenty of speech data from India (Vaani project) in the amount of over 154 hours, in Bangladesh (RBVD) there is only about 50 minutes of data (1,067 samples). In addition, RBVD samples contain significantly less data in average (samples of 2 seconds compared to those of 5 seconds in Vaani) and need to be upsized. Moreover, problems with corrupted `.mp3` header files and demographic imbalances (such as the lack of age-related information or limited range of women in the RBVD dataset) have to be taken into account. Future attempts should therefore be made to collect the gold standard acoustic corpus with equal data and distribution of demographic parameters between these two countries. Despite being an optimal but difficult solution, future research might try developing a data mining pipeline to collect dialectical speech data from YouTube.

- **Model Architecture and Training Enhancements:** Given the relative scarcity of data, it was not possible to tune the inner layers of the XLS-R model, limiting tuning to a classification layer, which is learned through a weighted sum of the frozen layers. Techniques of calculating the importance of the layer, like Fisher Information Matrix, were also skipped due to computational limits. As a result of these decisions, the model heavily relied on semantic "lexical shortcuts" and individual acoustic transients. This is why future architectures need to make sure that the phonetic and prosodic features are extracted from the audio properly. In other words, future architectures should focus on optimizing the training of the intermediate transformer layers (responsible for phonetic modeling) without touching the upper semantic layers, which need to remain frozen so that there isn't any vocabulary bias.

Finally, combining temporal architectures (like one-dimensional CNNs and recurrent layers) between the transformer and classification layer will help extract long-range dialectal prosody effectively.

- **Evaluation Methodology and Domain Application:** The primary problem with the methodology was in the process of data augmentation, where techniques such as adding background noise and pitch-shifting were used to both the training and testing datasets. This means that the results presented in the paper are based on performance against a simulated distribution and not the actual distribution. Future experiments should ensure rigorous data sanitation to ensure complete segregation of testing data from simulated data to get real-world performance estimates for generalization capabilities. After proper evaluation of the experiment and generation of an effective model, the next step will involve customizing the generated model for practical applications. For example, a model trained on noisy data can be used in border security and defense communications, while clean models can facilitate inclusive AI services for rural areas.

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